

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250275858

Kind Code

A1

Publication Date

September 04, 2025

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MACHINE-LEARNED MOVEMENT DETERMINATION BASED ON INTENT IDENTIFICATION

Abstract

A mobility augmentation system monitors data representative of a user's motor intent and augments the user's mobility based on the monitored motor intent data. A machine-learned model is trained to identify an intended movement based on the monitored motor intent data. The machine-learned model may be trained based on generalized or specific motor intent data (e.g., user-specific motor intent data). A machine-learned model initially trained on generalized motor intent data may be re-trained on user-specific motor intent data such that the machine-learned model is optimized to the movements of the user. The system uses the machine-learned model to identify a difference between the user's monitored movement and target movement signals. Based on the identified difference, the system determines actuation signals to augment the user's movement. The actuation signals determined can be an adjustment to a currently applied actuation such that the system optimizes the actuation strategy during application.

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Family ID: 81853558

Appl. No.: 19/209615

Filed: May 15, 2025

Related U.S. Application Data

parent US continuation 18602923 20240312 parent-grant-document US 12329661 child US 19209615

parent US continuation 17311058 20210604 parent-grant-document US 12102746 child US 18602923

Publication Classification

Int. Cl.: **A61F2/58** (20060101); **G06F3/01** (20060101); **G06F3/0346** (20130101); **G06N20/00** (20190101)

U.S. Cl.:

CPC **A61F2/583** (20130101); **G06F3/015** (20130101); **G06F3/016** (20130101); **G06F3/0346** (20130101); **G06N20/00** (20190101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 18/602,923, filed Mar. 12, 2024, which is a continuation of U.S. application Ser. No. 17/113,058, now U.S. Pat. No. 11,957,605, filed Dec. 6, 2020, which is incorporated by reference in its entirety. This application is also related to U.S. application Ser. No. 17/113,059, filed Dec. 6, 2020, now U.S. Pat. No. 12,005,573, which is incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to a mobility augmentation system, and more specifically to using movement intent to optimize and personalize mobility augmentation.

BACKGROUND

[0003] Hundreds of millions of people live with a disability. In the United States, approximately 14% of adults with a disability have a mobility disability that causes a person serious difficulty walking or climbing stairs, according to a 2019 study by the Centers for Disease Control and Prevention. Conventional methods of assisting mobility, such as crutches, canes, and wheelchairs are not sufficient to enable individuals to achieve full independence and mobility.

[0004] While technology such as exoskeletons and therapeutics such as functional electrical stimulation (FES) mark improvements over conventional methods for increasing mobility, such technology suffers from similar limitations as conventional crutches, canes, and wheelchairs: they are not personalized to the user and fail to optimize mobility augmentation based on available information (e.g., the user's movement). Accordingly, existing technology may be improved by personalizing and optimizing mobility augmentation.

SUMMARY

[0005] The mobility augmentation system described herein implements machine learning and control mechanisms to personalize and optimize mobility augmentation. The system monitors the user for various data associated with movement such as muscle electroactivity (i.e., muscle firing), kinematics, and kinetics. By monitoring muscle electroactivity, the system can determine what movement the user intends to make before the user makes it. This is an improvement over conventional systems that merely use inertial measurement unit (IMU) data indicative of a user's current movement to determine whether to augment movement. This reactive approach taken by conventional systems cannot assist users impacted with neuro-atypical motor function before they attempt a neurotypical movement.

[0006] The mobility augmentation system described herein determines intended movements using a machine-learned model that identifies an intended movement or movement prediction based on monitored data such as muscle electroactivity, kinematics, and kinetics. The machine-learned model can be trained on generalized movement data collected across a population of users or on data associated with a particular user's movement, which fine tunes its movement predictions for

that user and enables personalized mobility augmentation. The system uses the movement prediction to determine mobility augmentation or an actuation strategy. This actuation strategy can be further personalized to a user.

[0007] Once the system applies an actuation strategy, it further monitors the user's movement to gauge how successful the applied actuation strategy was. By comparing the monitored movement and target movement associated with the user's intention, the system can re-train the machine-learned model (e.g., when the actuation strategy was appropriate for the identified intended movement). In this way, the system further personalizes the mobility augmentation to the user. Furthermore, after comparing the monitored and target movements, the system may adjust the actuation strategy to minimize subsequent differences between monitored and target movements. Accordingly, the system optimizes actuation strategies by monitoring the user after applying actuation.

[0008] In one embodiment, the mobility augmentation system collects a first set of motor intent data of one or more users from a database. Examples of motor intent data include electromyography (EMG) data, IMU data, foot plantar pressure signals, or a combination thereof. The system labels the first set of motor intent data with an intent label representative of intended motion characterized by the first set of motor intent data. For example, the motor intent data may be labeled with an intent label that indicates that a user intended to take a step forward or lift a toe. The system creates, based on the labeled first set of motor data, a first training set to train a machine learning model. The machine learning model is configured to output, based on monitored motor intent data, a movement prediction corresponding to likely motion characterized by the monitored motor intent data. The monitored motor intent data may be captured by sensors located at various areas on a user's body (e.g., a target user distinct from the users that contributed to the first set of motor intent data). The system creates a second training set based on the movement prediction and a second set of motor intent data corresponding to movement signals of the target user. The machine learning model is re-trained using the second training set such that it is customized to the motions of the target user.

[0009] In another embodiment, the mobility augmentation system applies a machine learning model to identify an intended movement of a user. The system monitors movement signals representative of the user's movement. The machine learning model is used to determine an intended movement of the user based on motor intent data received from the user (e.g., via sensors located on the user's body). Using the intended movement determination, the system identifies a difference between the movement signals and target movement signals. The target movement signals may be representative of the user's intended movement. For example, the system determines that the user intends to stand up, and the target movement signals include kinematic, kinetic, EMG signals, or any combination thereof associated with neurotypical standing movement. The system determines actuation signals based on this identified difference between monitored and target movement signals. For example, parameters for an FES stimulation for assisting a user in standing may be determined based on how close the user's monitored movement is to the target movement. The system then applies the determined actuation signals to one or more augmentation devices, or "mobility augmentation devices," worn by the user. For example, the determined FES stimulation may be applied through actuation electrodes located at each of the devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a block diagram of a system environment in which a mobility augmentation device operates, in accordance with at least one embodiment.

[0011] FIG. 2 is a block diagram of the mobility augmentation system of FIG. 1, in accordance

with at least one embodiment.

[0012] FIG. 3 is a block diagram of a feedback loop for optimizing actuation, in accordance with at least one embodiment.

[0013] FIG. 4 depicts a visualization of electrical activity of muscles involved in a user's gait, in accordance with at least one embodiment.

[0014] FIG. 5 depicts a visualization of kinematics of muscles involved in a user's gait, in accordance with at least one embodiment.

[0015] FIG. 6 depicts a visualization of foot pressure reported by the mobility augmentation system during a user's gait, in accordance with at least one embodiment.

[0016] FIGS. 7A and 7B illustrate a wearable form factor of a mobility augmentation device for a user's legs, in accordance with at least one embodiment.

[0017] FIGS. 8A and 8B illustrate wearable, modular form factors of a mobility augmentation device, in accordance with at least one embodiment.

[0018] FIGS. 9A and 9B illustrate a wearable form factor of a mobility augmentation device for a user's foot, in accordance with at least one embodiment.

[0019] FIG. 10 is a flowchart illustrating a process for applying actuation signals to an augmentation device worn by a user, in accordance with at least one embodiment.

[0020] FIG. 11 is a flowchart illustrating a process for training a machine learning model configured to output a movement prediction based on monitored motor intent data, in accordance with at least one embodiment.

[0021] FIG. 12 shows an experimental finding of action prediction using inertial measurement unit data.

[0022] FIG. 13 shows an experimental finding of action prediction using electromyography data.

[0023] FIG. 14 shows an experimental finding of knee and hip kinematics augmented with functional electrical stimulation.

[0024] The figures depict various embodiments for purposes of illustration only. One skilled in the art will readily recognize from the following discussion that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein.

DETAILED DESCRIPTION

System Architecture

[0025] FIG. 1 is a block diagram of a system environment **100** in which mobility augmentation devices operate. The system environment **100** shown by FIG. 1 includes a mobility management system **110**, mobility augmentation devices **120a** and **120b**, a mobility augmentation system **130**, a training database **140**, a remote therapy system **150**, and a network **160**. The system environment **100** may have alternative configurations than shown in FIG. 1, including for example different, fewer, or additional components. For example, an additional mobility augmentation device may be communicatively coupled with the network **160** to the mobility management system **110**. In another example, the remote therapy system **150** may be omitted from the system environment **100** without compromising the functionality of the mobility augmentation devices **120a** and **120b**.

[0026] The system environment **100** enables the augmentation devices **120a** and **120b** to increase the mobility of its users. In a first example, a child with Cerebral palsy wears the mobility augmentation devices **120a** and **120b** at different locations on his body to step more naturally through his gait. In a second example, a patient suffering from advanced Parkinson's Disease (e.g., experiencing motor symptoms including tremors and freezing) wears one or more of the mobility augmentation devices **120a** and **120b** to restore his ability to perform otherwise difficult activities (e.g., standing up from a chair).

[0027] While the previous examples involved users with neuro-atypical movement, the system environment **100** may also help users with neurotypical movement increase or maintain their mobility. In a third example, a first responder carrying an injured person from a wreckage site

wears the mobility augmentation device **120a** to maintain her stamina or increase her speed to save lives. In a fourth example, a dancer wears the mobility augmentation device **120a** to improve her form while performing an arabesque. In some embodiments, the system environment **100** may assist medical professionals in diagnosing or treating their patients. In a fifth example, a clinician wears the mobility augmentation device **120a** to experience the muscle stimulation the device **120a** delivers to the clinician's patient. This fifth example may improve a clinician's understanding of what the patient's muscle sequencing is like (e.g., during an educational course, physical training, or therapy) as compared to the patient try to explain what the patient's muscle sequencing feels like during movement.

[0028] The mobility augmentation devices **120a** and **120b** increase the mobility of its users by monitoring for a user's intended movement and modifying or augmenting the movement by applying actuation signals. The devices **120a** and **120b** may determine the actuation signals using target movement signals (e.g., neurotypical movement). The devices **120a** and **120b** may be worn at various locations on the body of the user to monitor for the user's intended movement or motor intent data. For example, the devices **120a** and **120b** may use electromyography to monitor the electrical activity of the user's muscles. From the monitored electrical activity, the devices **120a** and **120b** may determine the user's intended movement using one or more machine learning models trained to identify a likely movement the user intends to make. The devices **120a** and **120b** may determine actuation signals to apply based on the identified intention. After the devices **120a** and **120b** determine the actuation signals to apply, the devices **120a** and **120b** apply the actuation signals to the various locations on the body of the user.

[0029] The mobility augmentation devices **120a** and **120b** enable both personalization and optimization of mobility augmentation for their users. One way in which the devices **120a** and **120b** personalize mobility augmentation is by using movement data collected from a given user's movement history to train a user-specific machine learning model that is used in determining the actuation signals for augmenting the user's subsequent movements. The devices **120a** and **120b** may optimize mobility augmentation by measuring the success of the actuation signals in real time and varying the subsequently applied actuation signals based on the measurement. Another way that the devices **120a** and **120b** optimize mobility augmentation may be through initial and continuous electric or electronic calibrations before and during collection of a user's movement signals used for determining the user's intended movement. Personalization and optimization will be described in further detail throughout the description of the mobility augmentation system **130** in FIG. 2.

[0030] As referred to herein, a “movement signal” is a signal representative of a user's movement. For example, the movement signal may be a kinematic, kinetic, foot pressure, or electrical activity signal, or any suitable combination thereof. Movement data may be a collection of one or more movement signals. As referred to herein, “motor intent data” is data representing a user's intended movement before or during the movement. For example, electrical activity of a user's muscle monitored by the mobility augmentation device **120a** via electromyography represents the user's intention to take a step backwards before he takes the step. As referred to herein, a “target movement signal” is a signal representative of desired movement. For example, kinematic signals collected across a population with neurotypical gaits may be averaged to create a target movement signal for a gait.

[0031] As referred to herein, an “actuation signal” is a signal carrying stimulation or instructions to actuate stimulation. For example, an actuation signal can be an FES signal applied via an electrode or instructions to reverse contraction motors of a prosthetic hand. An actuation strategy may be a particular delivery of one or more actuation signals to one or more portions of a user's body to achieve a target movement signal. The terms “movement” and “motion” may be used interchangeably herein to refer to a body's change in position over time. The terms “target user” and “user” may be used interchangeably herein to refer to a wearer of a mobility augmentation

device unless another meaning is apparent from the context. Although “users” described herein are human users, the systems and methods described herein may be similarly applied to augmenting movement for animals as well.

[0032] The mobility management system **110** monitors and processes data from the mobility augmentation devices **120a** and **120b**. The data received from the devices **120a** and **120b** may include motor intent data, movement data, and applied actuation strategies. This data may be used to generate new actuation strategies or modify existing actuation strategies. The mobility management system **110** may provide actuation strategies to the augmentation devices **120a** and **120b**. For example, during initial use and if the augmentation device **120a** has not already been customized to its user, the mobility management system **110** may provide an actuation strategy with target movement signals representative of a neurotypical gait that has been generalized from the neurotypical gaits of a group of people. The mobility management system **110** may be hosted on a server or computing device (e.g., a smartphone) that communicates with the mobility augmentation devices **120a** and **120b** via the network **160**.

[0033] In some embodiments, the mobility management system **110** trains and applies one or more machine learning models configured to determine a user's intended movement based on monitored motor intent data. The mobility management system **110** may maintain machine learning models in addition to or alternative to the mobility augmentation devices **120a** and **120b** maintaining the models. In one embodiment, the mobility management system **110** trains the models based on motor intent data collected by the devices **120a** and **120b**. The devices **120a** and **120b** send, via the network **160**, motor intent data to the mobility management system **110** and leverage the trained machine learning models to receive, from the mobility management system **110**, a likely intended movement determined by the one or more models. The mobility management system **110** may maintain models that are generalized to movement across a population or customized to a particular user, movement type, any suitable phenotypic trait, or a combination thereof. The training and application of machine learning models used for augmenting mobility is further described in the description of FIG. 2.

[0034] Mobility augmentation devices **120a** and **120b** augment a user's movement by monitoring intended movement data and applying actuation signals determined based on a target movement signal. The devices **120a** and **120b** may optimize the augmented movement by implementing a control system that adjusts the applied actuation signals over time to minimize a difference between the target movement signal and monitored movement signals. The device **120a** includes a controller **121**, actuators **122**, sensors **123**, and a mobility augmentation system **130**. The device **120b** includes similar hardware and software components as in the device **120a**.

[0035] The devices **120a** and **120b** may have alternative configurations than shown in FIG. 1, including for example different, fewer, or additional components. For example, the devices **120a** and **120b** may include additional components such as one or more processors (e.g., a general purpose processor and digital signal processor), wireless communications circuitry for enabling communication via the network **160**, signal generators for generating functional electrical stimulation, an input interface (e.g., a keyboard or a microphone), an output interface (e.g., a speaker or a display), supplemental memory (e.g., an SD memory card), or a power source. Additionally, although the controller **121** is depicted as separate from the mobility augmentation system **130**, the mobility augmentation system **130** may perform the functionality of the controller **121** (i.e., the controller **121** is encompassed within the system **130**).

[0036] The mobility augmentation devices **120a** and **120b** may have various, wearable form factors such as exoskeletons, modular electrode straps, leggings, foot pressure beds, any wearable form factor suitable for targeting a particular muscle group on a user's body, or a combination thereof. For example, the device **120a** may be a legging that is worn underneath regular attire and is equipped with the sensors **123** and actuators **122** for performing the mobility augmentation described herein. Examples of form factors are illustrated in and further described in the

descriptions of FIGS. 7-9.

[0037] The controller **121** optimizes the actuation strategy implemented by the mobility augmentation device **120a** to minimize the difference between a target movement signal and a measured movement signal. In some embodiments, the sensors **123** measure the user's movement while the actuation strategy is applied by the actuators **122**. The controller **121** compares the measured movement signals to target movement signals. Based on the comparison, the controller **121** modifies the actuation strategy. A feedback process implemented by controller **121** is described in further detail in the description of FIG. 3.

[0038] The actuators **122** apply actuation signals to the user. The actuators **122** may have varying types including electric, mechanic, haptic, audio, visual, pneumatic, hydraulic, or any combination thereof. The form factor of the mobility augmentation devices **120a** and **120b** may determine the type of the actuators **122**. For example, a mobility augmentation device having an exoskeleton form factor may include a combination of pneumatic and hydraulic actuators, where applying an actuation signal involves actuating a limb of the exoskeleton via one of the pneumatic and hydraulic actuators. The mobility augmentation system **130** may determine the combination of actuation types to use depending on the user such that the mobility augmentation is personalized to the user. For example, the mobility augmentation system **130** may determine that the user's gait is maximally optimized when haptic actuation is applied instead of visual or audio actuation by monitoring the user's augmented movement and determining associations between augmented movement and the actuation type.

[0039] The actuation signals may be determined by the mobility augmentation system **130** or manually specified by an operator (e.g., a physical therapist) or the user through an input interface on the devices **120a** and **120b**. In some embodiments, the actuation signal is an FES signal characterized by a frequency, a pulse duration, and an amplitude (e.g., a value of current in milliamperes). The mobility augmentation system **130** may determine if and how the actuation signal changes over time. For example, the FES signal may have a first frequency, a first pulse duration, and a first amplitude for a first period of time, and then a second frequency, a second pulse duration, and a second amplitude for a second period of time following the first period of time.

[0040] The actuators **122** may execute a variety of actuation types. Examples of actuation types include manually triggered actuation, amplification, contralateral replay, body-to-body coaching, templated sequencing, and responsive optimization. The user of the mobility augmentation device **120a** or a third party may manually trigger an actuation via the actuators **122**. For example, a user can instruct the device **120a** to generate FES stimulation and apply via the actuators **122**). Other actuation types may be automatically determined by the mobility augmentation devices **120a** or **120b**, and are described both in the following paragraphs and in the description of FIGS. 2-3.

[0041] During amplification, the actuators **122** amplify a user's existing movements. For example, the actuators **122** use FES to stimulate muscles involved in closing a fist as the user is closing his fist to grasp an object. In contrast, amplification may not be applicable to a user with tremors due to Parkinson's Disease, as amplification would worsen his condition. Amplification may be accomplished by sensing electroactivity from a particular muscle and subsequently triggering FES in the same muscle. In some embodiments, amplification is used to calibrate sensor position and the intensity of stimulation. Calibration is further described in the description of FIG. 2.

[0042] Contralateral replay may be applicable for users who have an injury or weakness on one side of their body and not the other (e.g., users who have suffered from a stroke). The mobility augmentation devices **120a** and **120b** may enable a user to leverage the user's stronger side of their body to help train the movements of the weaker side. The device **120a** may be located at a first location on the user's body and the device **120b** is located at a second location that is mirrored across the sagittal plane of the user's body. For example, the sensors **123** located on both the left and right leg may be used to monitor the user's motor intent data and movement data, while the

actuators **122** on the weaker, right leg are used to apply the actuation. The sensors **123** capture the muscle firing or kinematic patterns from the left leg and the mobility augmentation system **130** determines, based on these patterns from the left leg, the actuation to apply via the actuators **122** on the right leg.

[0043] Body-to-body coaching involves the participation of an operator or third-party user to produce target movement signals to coach the user wearing the mobility augmentation devices **120a** and **120b**. In some embodiments, the operator is equipped with a mobility augmentation device having sensors to measure the operator's movement or intended movement prior to movement (e.g., EMG signals). The operator's mobility augmentation device provides the measured data or identified movement associated with the measured data over the network **160** for the devices **120a** or **120b** to receive. For example, the operator's mobility augmentation device may send data collected from IMU sensors or may process the data to determine the operator is lifting his right foot and send that determination to the devices **120a** or **120b**. The devices **120a** or **120b** may then reproduce the operator's motion on the user. For example, the devices **120a** receives the indication that the operator lifted his right foot and the mobility augmentation system **130** determines the appropriate actuation strategy to lift the user's right foot.

[0044] The actuators **122** may implement templated sequencing, applying actuation signals based on templates associated with particular movements. The mobility management system **110** may collect data across neurotypical populations to form the templates. In some embodiments, the templates include a sequence of mappings between specific events in a movement that corresponds to the beginning of ending of a specific muscle firing. The sequence of mappings are organized chronologically in the order in which the events occur in a movement. For example, a template for a gait can include events associated with a foot leaving the ground followed by events associated with a leg lift, swing, and finally, the heel of the foot striking the ground to complete the gait. An example visualization of a template is described in the description of FIG. 4.

[0045] Responsive optimization may be performed in addition to any one of the aforementioned actuation types. In particular, the mobility augmentation devices **120a** and **120b** can gauge the success of the actuation strategy applied by the actuators **122** and adjust subsequent actuation to minimize the difference between target and measured movement. The adjusted actuation is then applied by the actuators **122**. Responsive optimization is further described in the description of FIG. 2.

[0046] The sensors **123** monitor the user for intended movement data and movement data. The sensors **123** may be one or more of a microelectromechanical systems (MEMS) device, IMU, sensing electrodes, pressure sensor bed, any suitable device for measuring kinetic or electrical signals produced by a muscle, or a combination thereof. The sensors **123** may be located at various locations on the user's body. For example, a pressure sensor bed may be placed in the user's right shoe to measure the user's right foot pressure as he completes a gait. A set of sensing electrodes may be placed at the shank of the user's right leg to measure the intended movement data before and during the gait. The sensors **123** may be communicatively coupled to the controller **121** or a processor of the mobility augmentation device **120a** to provide the monitored data to determine or optimize actuation signals applied by the actuators **122**.

[0047] The mobility augmentation system **130** determines an intended movement of the user and augments the movement associated with the user's intention. In some embodiments, the mobility augmentation system **130** receives intended movement data from the sensors **123** and preprocesses the data before applying one or more machine learning models to the preprocessed intended movement data. The one or more machine learning models are configured to determine the user's intended movement such as standing up or stepping forward. Once the intended movement is determined, the mobility augmentation system **130** determines a difference between the user's current movement and the intended movement. The mobility augmentation system **130** may determine this difference by identifying target movement signals representative of the intended

movement and comparing the target movement signals to movement signals included in the movement data received from the sensors **123**. The mobility augmentation system **130** determines actuation signals to apply based on the determined difference. For example, the mobility augmentation system **130** may determine the amplitude of FES to the user's legs as proportional to the difference between a movement signal and a target movement signal (i.e., the smaller the error, the less stimulation needed to augment the user's movement to achieve the desired movement). The mobility augmentation system **130** is further described in the description of FIG. 2.

[0048] The training database **140** includes various data for training machine learning models of the mobility augmentation system **130**. The data stored in the training database **140** may include labeled or unlabeled motor intent data and associated movement data (i.e., the measured movement associated with the intention), labels associated with movements, or templates associated with sequences of muscle firings for given movements. The mobility management system **110** or the mobility augmentation devices **120a** and **120b** may access the stored data to train machine learning models. The mobility augmentation devices **120a** and **120b** may provide their measured data to the training database **140**. The provided data may be organized in a data structure including the measured data, biographical information identifying the user and phenotypic traits, and a label identifying the intended movement.

[0049] The remote therapy system **150** enables a third party (e.g., a medical professional or athletic coach) to monitor the user's movement and analyze the information to further augment the user's movement. For example, a physician uses the remote therapy system **150** to monitor his patient's movement and adjust an actuation strategy upon identifying that the patient's movement is not improving under the current actuation strategy. In particular, the electrical activity data of a patient's muscles measured by mobility augmentation devices **120a** and **120b** help a physician diagnose the needs of the patient more accurately than related systems of art that rely solely upon IMU data to trigger movement augmentation. The remote therapy system **150** may be a software module that the third party may execute on a computing device (e.g., a smartphone). In some embodiments, the remote therapy system **150** is a standalone device that may be communicatively coupled to a mobility augmentation device to manually adjust or generate actuation signals used to augment the user's motion (e.g., overriding the mobility augmentation system **130**). The remote therapy system **150** may include an input interface for the third party to specify parameters of an actuation signal (e.g., the amplitude and frequency of FES signals) and when to apply them.

[0050] The remote therapy system **150** may provide actuation strategies to be applied by the mobility augmentation system **130**. In some embodiments, a user of the remote therapy system **150** (e.g., a therapist) may specify when to apply stimulation and, for an array of mobility augmentation devices worn by a patient, which of the devices to apply stimulation to. For example, the therapist may define where and when to stimulate the patient's gait based on a video camera of the sensors **123** that capture the patient's gait. The therapist-specified actuation strategy may be communicated from the remote therapy system **150** to the mobility augmentation devices **120a** and **120b** over the network **160**.

[0051] The network **160** may serve to communicatively couple the mobility management system **110**, the mobility augmentation devices **120a** and **120b**, the training database **140**, and the remote therapy system **150**. For example, the mobility augmentation device **120a** and the remote therapy system **150** are configured to communicate via the network **160**. In some embodiments, the network **160** includes any combination of local area and/or wide area networks, using wired and/or wireless communication systems. The network **160** may use standard communications technologies and/or protocols. For example, the network **160** includes communication links using technologies such as Ethernet, 802.11, worldwide interoperability for microwave access (WiMAX), 3G, 4G, 5G, code division multiple access (CDMA), digital subscriber line (DSL), etc. Examples of networking protocols used for communicating via the network **160** include multiprotocol label switching (MPLS), transmission control protocol/Internet protocol (TCP/IP), hypertext transport protocol

(HTTP), simple mail transfer protocol (SMTP), and file transfer protocol (FTP). Data exchanged over the network may be represented using any suitable format, such as hypertext markup language (HTML) or extensible markup language (XML). In some embodiments, all or some of the communication links of the network **160** may be encrypted using any suitable technique or techniques.

[0052] Although the components of the system environment **100** are shown as connected over the network **160**, one or more components may function without being connected to the network **160**. For example, the augmentation devices **120a** and **120b** may function offline when the devices **120a** and **120b** are not able to connect to the network **160**. When the devices **120a** and **120b** are able to reconnect to the network **160**, they may upload monitored data or actuation results to the mobility management system **110** or the remote therapy system **150** via the network **160**.

Mobility Augmentation System

[0053] FIG. **2** is a block diagram of the mobility augmentation system **130** of FIG. **1**. The mobility augmentation system **130** includes local databases such as an intent label database **200**, a template database **210**, and a personalized actuation database **220**. The mobility augmentation system **130** includes software modules such as an intent label determination module **230**, a preprocessing module **240**, a calibration module **250**, an actuation determination module **260**, a responsive optimization module **261**, and an activity prediction module **270**, and a machine learning model training engine **271**. The activity prediction module **270** further includes machine learning models such as a general movement model **272**, a task-specific model **273**, and a user-specific model **274**. The mobility augmentation system **130** may have alternative configurations than shown in FIG. **2**, including different, fewer, or additional components. For example, one or more of the databases **200**, **210**, or **220** may be stored remotely rather than on the mobility augmentation device (e.g., contents stored in the training database **140**) and may be accessible through the network **160**. In another example, an additional report generation module may generate a report of the applied actuation and the monitored movement data associated with the actuation and provide the report to the remote therapy system **150**.

[0054] The intent label database **200** stores labeled motor intent data. The machine learning model training engine **271** may use the data stored in the intent label database **200** to train one or more of the machine learning models used by the activity prediction module **270**. The data stored in the intent label database **200** may be user-specified, determined by the mobility augmentation system **130**, or a combination thereof. In some embodiments, a human administrator of the mobility management system **110** manually labels motor intent data and provides the labeled data for storage in the intent label database **200** via the network **160**. Additionally or alternatively, the data stored in the intent label database **200** is motor intent data measured by the sensors **123** and labeled by the intent label determination module **230**.

[0055] Label types may be associated with specific muscles or motions of muscles. The motor intent data within the intent label database **200** may be labeled according to varying degrees of specificity. Examples of general movement labels include “jump,” “stand,” “walk,” and “step backward.” Activity-specific movement labels can include “putt,” “forehand swing,” and “arabesque.” Muscle-specific movement labels can include “knee joint extension” and “hip flexion.” One or more labels may be applied to motor intent data. For example, a set of motor intent data may be labeled with “walk,” “knee joint extension,” and “hip flexion.”

[0056] The template database **210** stores templates for the mobility augmentation devices **120a** and **120b** to implement templated sequencing. Template types may be associated with specific phenotypic traits or activities. For example, the template database **210** may store a template for an activity such as jumping. In another example, the template database **210** may store a template for users of a particular weight or age range. A template may include target movement signals such as target kinematic, foot pressure, or kinetic signals. In some embodiments, the templates are pre-determined. For example, the templates can be provided by the mobility management system **110**

over the network **160**.

[0057] In some embodiments, the mobility management system **110** determines the templates and provides the determined templates to the mobility augmentation devices **120a** and **120b**. The mobility management system **110** may access a particular set of movement data to generate a corresponding template. For example, the mobility system **110** may receive movement data from the augmentation device **120a** and **120b** to create templates that are user-specific. Additionally, the mobility management system **110** may identify particular phenotypic traits across users and generate phenotype-specific templates based on those users' movement data. For example, the mobility management system **110** identifies users who are within the ages 60-65 years old and analyzes their movement data to identify templates for general movements such as walking, standing, sitting, or grasping. To generate activity-specific templates, the mobility management system **110** accesses movement data collected across a neurotypical population performing a particular activity, such as a jump.

[0058] The mobility management system **110** then analyzes the accessed movement data for the start and end of specific muscle firing events such as the start and end of the right shank crossing a given degree of flexion. The mobility management system **110** associates or maps a start or end of the muscle firing events to respective triggers or actuation signals. For example, the end of the right shank crossing 30 degrees of flexion in a negative direction may be associated with an actuation signal applied to an actuator located at the right gastrocnemius. The mobility management system **110** may then aggregate these mappings into a template, which the system **110** provides to the mobility augmentation devices **120a** and **120b**.

[0059] The personalized actuation database **220** stores user-specific modifications to generalized actuation strategies. The responsive optimization module **261** may determine adjustments to generalized actuation strategies provided by the mobility management system **110**. The responsive optimization module **261** may store successful adjustments (i.e., updated actuation signals that minimized the difference between target and measured movement) in the personalized actuation database **220**. The user-specific modifications may include adjustments to actuation strategies with predefined actuation parameters such as an FES signal amplitude, a timing of signal application, actuators used (i.e., in an arrayed actuation configuration where an array of actuators is used to augment movement), an actuation type, any suitable characteristic defining the actuation strategy, or a combination thereof. Additionally, the personalized actuation database **220** may store user-specific calibration settings determined by the calibration module **250**.

[0060] The intent label determination module **230** uses computer vision to derive labels from measured motor intent data. The intent label determination module **230** may learn the appropriate label or labels for measured motor intent data via reinforcement learning. The intent label determination module **230** may be rewarded for its label determination after measured movement data associated with the applied actuation and the determined label results in minimal differences between the measured movement and the target movement. The intent label determination module **230** provides correctly labeled motor intent data to the intent label database **200**. If the label is incorrect, the motor intent data will not be stored into the intent label database **200** with the incorrect label.

[0061] The intent label determination module **230** may use a combination of one or more of computer vision, foot pressure measurements, IMU data, EMG data, and pre-defined labels to determine a label. For example, the intent label determination module **230** may determine a label for motor intent data based on a weighted combination of a label determined from reinforcement learning and manually labeled motor intent data, where a greater weight is placed upon similarities between monitored motor intent data and the manually labeled motor intent data. In some embodiments, the intent label determination module **230** includes a machine learning model using an unsupervised learning algorithm to identify reoccurring patterns of motor intent data within different sets (e.g., taken at different times or from different users) and recognize a potential label

for the data.

[0062] The preprocessing module **240** may process the monitored motor intent data or movement data for use by other modules of the mobility augmentation system **130**. The module **240** may align data received from multiple sensors in the temporal domain. For example, the module **240** aligns EMG and IMU signals measured by the sensors **123** on both mobility augmentation devices **120a** and **120b** such that the system **130** can determine, for a given time, the value of the EMG and IMU signals measured by both devices at that time. This alignment may be referred to herein as “time alignment.” The module **240** may align measured data by determining an alignment that meets or exceeds threshold correlation value with aligned data provided by the mobility management system **110**. The devices **120a** and **120b** may associate measured data with timestamps, which may be used to align the measured data when aggregated for use by the system **130**.

[0063] The preprocessing module **240** may apply various digital signal processing (DSP) techniques to the measured data. The DSP techniques include filtering, averaging, peak-finding, down-sampling, Fourier transforms, root mean square (RMS), any suitable DSP technique, or a combination thereof. In one example, the module **240** may merge (i.e., time align) multiple channels (e.g., 2 kilohertz bandwidth channels) of EMG, IMU, and force sensitive resistor (FSR) data. The module **240** can preprocess the channels of EMG data by applying a differential filter and averaging a particular window in time (e.g., 100 milliseconds) for each channel. The module **240** down-samples and normalized the filtered and averaged data for use by the activity prediction module **270** as input to a machine-learned model. Additionally, the module **240** can apply a biomechanical model (e.g., inverse kinematics) or a machine learned model (e.g., a neural network) to EMG data processed using one or more of the techniques described herein to determine kinematic data associated with the muscle's electroactivity.

[0064] The preprocessing module **240** may transform a target movement goal to target movement signals. The target movement goal may be specified by a third party (e.g., a therapist of the remote therapy system **150** or an administrator of the mobility management system **110**). Alternatively or additionally, the target movement goal may be specified by the user of the mobility augmentation devices **120a** and **120b**. For example, the activity prediction module **270** may determine, using measured EMG signals via the sensors **123**, a user intends to make a particular movement and sets this movement as the target movement goal. Target movement goals include walking with normative kinematics, symmetric kinematics among both a user's legs, symmetric EMG profiles among both the user's legs, toe-heel running, reducing the user's energy expenditure while performing an action (e.g., walking), reducing pain caused by osteoarthritis (e.g., knee joint loading), any suitable motion of a body characterizable by signals (e.g., EMG, kinetics, kinematics, foot pressure, etc.), or a combination thereof.

[0065] In some embodiments, the preprocessing module **240** transforms the target movement goal to target movement signals that share a domain with the signal received from the sensors **123**, referred to herein as the sensors' domain. The transformation of a movement goal into the sensors' domain is applicable in a feedback loop such as the loop depicted in FIG. 3. For example, a target movement goal of reducing knee joint loading may be transformed to various target movement signals in the sensors' domain. The various target movement signals may include a 60-degree flexion at 70% of the gait cycle or a specific rectus femoris EMG RMS profile. The preprocessing module **240** may perform the transformation using a predetermined mapping between the goals and signal. In some embodiments, the preprocessing module **240** implements an algorithm that determines likely target movement signals to achieve the target movement for the user. For example, the preprocessing module **240** may input parameters characterizing the user (e.g., phenotypic characteristics) into an algorithm to determine likely target movement signals.

[0066] The calibration module **250** optimizes the actuators **122** and sensors **123** to personalize the movement augmentation to the user. Initial calibrations may be applied before and during the monitoring taken by the sensors **123** or the actuation applied by the actuators **122**. In some

embodiments, the calibration module **250** optimizes electrical activity measurements taken by the sensors **123** and FES applied by the actuators **122** by performing impedance matching at the locations on the body where the sensors **123** and the actuators **122** are located. Additionally, the calibration module **250** may apply a set of calibrating actuation signals and measure the resulting movements via the sensors **123** to determine initial adjustments to generalized actuation strategies. For example, the calibration module **250** may weaken the amplitude or intensity of actuation signals applied to a user whose build is smaller than average for the user's age group and whose movement may be overcompensated when a generalized actuation strategy is applied. Measurements and adjustments made by the calibration module **250** may be stored within the personalized actuation database **220**. The stored data may be accessed by the machine learning model training engine **271** or the mobility management system **110** to train a personalized machine learning model.

[0067] The actuation determination module **260** determines, based on the intended movement predicted using the activity prediction module **270**, an actuation strategy to apply via the actuators **122**. To determine the actuation strategy to apply, the actuation determination module **260** may determine the type of actuation to apply and then determine a strategy of that actuation type. Additionally, the actuation determination module **260** further optimizes the determined strategy using the responsive optimization module **261**.

[0068] The actuation determination module **260** may determine a type of actuation to apply before determining the actuation strategy. Actuation types may include manual triggering, amplification, contralateral replay, body-to-body coaching, and templated sequencing. The actuation determination module **260** may receive a request or instructions from the user (e.g., using an input interface on the mobility augmentation device **120a** specifying the desired actuation type. For example, the module **260** may receive a user's request to apply a one-time FES signal at the actuators **123**, determining that the type of actuation to apply is of the manual triggering type. In some embodiments, the module **260** determines not to apply a requested manual triggering of movement augmentation. The module **260** may determine that the requested movement augmentation interferes or is unnecessary given the user's current posture or motion. For example, the module **260** receives a request from a third-party operator to trigger actuation that would assist the user in standing. The module **260** may leverage the sensors **123** and records of movement data to determine, based on the user's previous movements and current posture, that the user is currently standing and does not need assistance from the device **120a** or **120b** to stand.

[0069] If the mobility augmentation device **120a** or **120b** is a device that the user has not used before or is being used at a location the user has not previously applied movement augmentation to, the actuation determination module **260** may determine the actuation type is amplification. The module **260** may call upon the calibration module **250** to use amplification to gauge the user's sensitivity to actuation (e.g., FES signals or exoskeleton movement) and determine initial adjustments to default actuation strategies. Additionally or alternatively, the module **260** may receive a request from the user to activate amplification until instructed to deactivate, a user-specified period of time expires, or a threshold degree of success in movement augmentation is reached (e.g., the sensors **122** measure that the user's movement signal is within $\pm 10\%$ of the target movement signal).

[0070] The actuation determination module **260** may determine the actuation type is contralateral replay or body-to-body coaching in response to receiving instructions specifying a device from which target movement signals will be received from. In some embodiments, if the specified device is associated with the user, the module **260** determines that the actuation type is contralateral replay. For example, a hardware identifier of the mobility augmentation device **120b** is assigned to the user's account, managed by the mobility management system **110**, and the module **260** receives instructions that the mobility augmentation device **120a** is to receive target movement signals from the device **120b**. In some embodiments, if the specified device is associated with another user (e.g.,

a coach), the module **260** determines that the actuation type is body-to-body coaching. For example, a hardware identifier of the device **120b** is assigned to the coach's account, managed by the system **110**, and the module **260** receives instructions that the device **120a** is to receive target movement signals from the device **120b**.

[0071] The actuation determination module **260** may set templated sequencing as a default actuation type to be used when an alternative actuation type is not applicable. For example, when a user wearing both the mobility augmentation devices **120a** and **120b** has not specified to the module **260** that the desired actuation type is contralateral replay, the actuation determination module **260** may determine to use templated sequencing and an actuation strategy that follows a template stored within the template database **210**.

[0072] After determining an actuation type, the actuation determination module **260** determines an actuation strategy of that type to apply via the actuators **122**. The actuation strategies may be predefined by the mobility management system **110**. In some embodiments, the actuation strategies are stored locally at the mobility augmentation devices **120a** or **120b**. An actuation strategy may define a combination of one or more actuation signals, a timing of delivery of the signals, and for an arrayed orientation of multiple actuators coupled to the mobility augmentation system **130**, which actuators are used to apply the signals. Each actuation strategy may be mapped to a corresponding predicted movement. For example, the activity prediction module **270** determines, based on monitored motor intent data, that the user is likely intended to grasp an object and in response, the actuation determination module **260** determines an actuation strategy that is mapped to the predicted movement of grasping. The module **260** may access the personalized actuation database **220** to use a personalized actuation strategy in place of a default actuation strategy (e.g., provided by the mobility management system **110**) or access the database **220** to store modified actuation that was determined using the responsive optimization module **261**.

[0073] In a first example of applying an actuation strategy, the actuation determination module **260** follows a template. A machine-learned model of the activity prediction module **270** may determine a movement prediction. The module **260** determines a template associated with the determined movement prediction, the template including mappings of muscle firing events to actuation signal triggers. For example, one of the muscle firing events in the template corresponds to when the right shank crosses 30 degrees of flexion in the negative direction. The mapped actuation signal trigger may correspond to stimulating the right gastrocnemius muscle with an FES signal at 30% of the maximum FES intensity. The module **260** may then use the motor intent data or monitored movement data to identify muscle firing events in the template performed by the user and apply respective triggers.

[0074] In second and third examples of applying an actuation strategy, the actuation determination module **260** applies an actuation strategy that is not necessarily associated with a template. In the second example, the module **260** increases the tension of an ankle-foot orthosis, which may be either one of the mobility augmentation devices **120a** or **120b**, by a predefined amount (e.g., specified by the mobility management system **110**) in response to a machine-learned model of the activity prediction module **270** predicting an intended movement of “life from chair” with at least 30% confidence. As referred to herein, the terms “intended movement” and “movement prediction” may be used interchangeably to refer to an output of a machine-learned model configured to determine a likely movement based on motor intent data. In the third example, the module **260** reverses contraction motors of a prosthetic hand, which may be either one of the devices **120a** or **120b**, in response to a machine-learned model of the module **270** predicting an intended movement of “release grip” with at least 60% confidence.

[0075] The responsive optimization module **261** of the actuation determination module **260** tailors generalized actuation strategies for the user. After the actuators **122** apply the actuation strategy determined by the module **260**, the sensors **123** monitor the user's movements to determine additional actuation to achieve the target movement. In some embodiments, the module **261**

receives manually input feedback in addition to or as an alternative to using the sensors **123**. For example, the mobility augmentation devices **120a** and **120b** may have an input interface including a button for the user to indicate that the applied actuation strategy was uncomfortable or to stop the applied actuation strategy. The module **261** may use this manually input feedback to determine adjustments to existing actuation or additional actuations.

[0076] The responsive optimization module **261** may determine which additional actuations or adjustments contribute to improving the movement augmentation and store those actuations or adjustments in the personalized actuation database **220** for future application. For example, the module **261** determines, based on monitored movement data by the sensors **123**, that the additional actuation associated with a particular actuation strategy minimizes the difference over time between the measured movement and target movement. The module **261** then stores a record of the actuation in the personalized actuation database **220**, which may be accessed to apply when the particular actuation strategy is next applied. The responsive optimization module **261** may use statistical optimization, reinforcement learning, or a combination thereof.

[0077] In one example of applying statistical optimization, the responsive optimization module **261** may use curve fitting to optimize a particular actuation strategy. The module **261** calculates a fit of the measured movement data (e.g., kinematics) of the user to a desired model (e.g., a target movement signal obtained from contralateral movement or a template). The module **261** then uses the calculated fit to adjust parameters of the actuation strategy to achieve maximal fit. For example, the module **261** may adjust the timing at which actuation signals are applied, delaying or advancing the delivery of signals as compared to a predefined timing schedule described in an actuation strategy.

[0078] A reinforcement learning approach may be used to optimize the actuation strategies as they are applied. The responsive optimization module **261** may create a set of rewards and penalties around certain features in measured movement data. Machine-learned models of the activity prediction module **270** may identify intended movements in the measured movement data. The identified intended movements may be features that are desired or unwanted given the goals of the actuation strategies applied. For example, if a machine-learned model identifies toe dragging movement in measured movement data taken after an actuation strategy associated with taking a step is applied, the module **261** may create a penalty to the actuation strategy around the toe dragging. In another example, gait symmetry may be used by the module **261** as a metric for rewarding and optimizing the actuation strategy. A movement pattern of a right, less-impacted leg may be used as a reference for scoring the left, impacted leg. The module **261** may compare the gait taken by the movement patterns of the left and right legs and use the comparison to vary the timing of when actuation signals are applied to the left leg to obtain a match between the gaits. For example, the more symmetric the gaits appear, the less the module **261** may need to vary the firing time or the module **261** may decrease timing variations by smaller amounts with each successive variation.

[0079] The activity prediction module **270** maintains one or more machine-learned models for determining an intended movement based on monitored motor intent data. The machine-learned models may include the general movement model **272**, the task-specific model **273**, and the user-specific model **274**. The machine learning model training engine **271** of the module **270** trains the machine learning models to determine a user's intended movement using different types of data sets, where the data set type is associated with a degree to which a model is tailored to the user. For example, the training engine **271** may use a data set that is unique to an activity of jumping, and the model trained using the data set is tailored to determine that a user intends to jump. The machine learning models are configured to receive, as input, monitored motor intent data (e.g., EMG signals) and output a likely value of an IMU signal at a time in the future (e.g., a second ahead of the current EMG signals). The machine-learned models may also output a confidence score corresponding to the intended movement.

[0080] The machine learning model training engine **271** may train a machine learning model in multiple stages. In a first stage, the training engine **271** may use generalized motor intent data collected across one or more users (e.g., a neurotypical population) to train the machine learning model. The intent label determination module **230** may label the generalized motor intent data with an intent label representative of the intended motion characterized by the generalized motor intent data. The training engine **271** then creates a first training set based on the labeled generalized motor intent data. The training engine **271** trains a machine learning model, using the first training set, to determine a movement prediction. That is, the machine learning model is configured to receive, as an input, monitored motor intent data (e.g., from the sensors **123**), and output the movement prediction corresponding to the likely motion characterized by the monitored motor intent data. The likely motion may include a likely IMU data value at a time occurring after a time when the motor intent data is monitored.

[0081] In a second stage of training, the training engine **271** uses user-specific motor intent data collected by the sensors **123**. The machine learning model training engine **271** creates a second training set based on previously determined movement predictions and the user-specific motor intent data. The movement predictions, depending on the success of the actuation strategies applied based on the movement predictions, may serve as labels for the user-specific motor intent data. If a previously determined movement prediction resulted in successful movement augmentation, the training engine **271** may create the second training set that includes user-specific motor intent data labeled with the determined movement prediction. The training engine **271** then re-trains the machine learning model using the second training set such that the machine learning model is customized to the user's motions.

[0082] To create a training set, the machine learning model training engine **271** may determine one or more feature vectors associated with a combination of different muscles and the timing of their firing during the intended motion. For example, the training engine **271** may determine a feature vector characterizing muscle firing events associated with a certain degree of knee flexion and a toe off event during a gait cycle. In some embodiments, the training engine **271** may receive calibration data associated with calibration performed prior to actuation and the resulting movement affected by the actuation. The training engine **271** may use the calibration data in creating the training set such that the trained machine-learned model is further customized to the user's motions.

[0083] In some embodiments, the machine learning model resulting from the second stage of training is maintained as a separate machine learning model from the model resulting from the first stage of training. For example, the general movement model **272** is the model resulting from the first stage of training and the user-specific model **274** is the model resulting from the second stage of training. Additionally, the task-specific model **273** may be the model resulting from the second stage of training when the motor intent data used in the second stage is specific to a task performed by the user.

[0084] Machine learning models of the activity prediction module **270** may use various machine learning techniques such as linear support vector machine (linear SVM), boosting for other algorithms (e.g., AdaBoost), neural networks, logistic regression, naïve Bayes, memory-based learning, random forests, bagged trees, decision trees, boosted trees, boosted stumps, a supervised or unsupervised learning algorithm, or any suitable combination thereof. The machine learning models may have access to a broader set of features on which to train. The models may use physiological simulation as a component for determining a movement prediction associated with an optimal actuation strategy.

[0085] The general movement model **272** is trained by the machine learning model training engine **271** using motor intent data collected across a neurotypical population performing a variety of general movements. The general movements may include walking, standing (i.e., from a sitting position), sitting, ascending or descending steps, grasping, any suitable movement used in day-to-

day activity, or a combination thereof.

[0086] The task-specific model **273** is trained by the machine learning model training engine **271** using motor intent data collected across a neurotypical population performing a specific task or gesture. The specific task may be a single type of movement that the general movement model **272** is trained to identify (e.g., walking or a gait cycle associated with walking). In some embodiments, the specific task is unique to an activity such as the performing arts or sports. For example, the task-specific model **273** may be trained to identify when a dancer intends to perform an arabesque.

[0087] The user-specific model **274** is trained by the machine learning model training engine **271** using motor intent data collected from the sensors **123**. The model **274** may be obtained by re-training the general movement model **272** using monitored motor intent data collected from the user of the augmentation device **120a**. Because the model **274** is trained on user-specific motor intent data, the model **274** enables the mobility augmentation system **130** to be personalized to the user and improve its accuracy in identifying movements that the user intends.

[0088] Although the activity prediction module **270** is depicted as being a component of the mobility augmentation device **120a**, the mobility management system **110** may have a similar functionality. The mobility management system **110** may create training sets based on monitored motor intent data associated with different users or different tasks. The system **110** may train a model similar to the task-specific model **273** or user-specific model **274** and provide the trained models to the mobility augmentation devices **120a** and **120b**. The system **110** may apply a machine-learned model to determine a movement prediction based on monitored motor intent data provided by the device **120a** or **120b** and provide the prediction to the mobility system **130** via the network **160**. In one example, motor intent data measured by the sensors **123** are stored in a local SD memory card at the device **120a**, the mobility system **130** uploads data from the SD card to a cloud server (e.g., the training database **140**), and the system **110** accesses the training database **140** to re-train and finetune a machine-learned model. The system **110** may be hosted on a remote computing device such as a smartphone.

[0089] FIG. **3** is a block diagram of a feedback loop **300** for optimizing actuation. The feedback loop **300** is a closed-loop system that minimizes differences between a user's movement and target movement. The mobility augmentation system **130** may perform the feedback loop **300**. The feedback loop **300** includes a preprocessor **310**, a summing point **320**, a controller **330**, an FES generator **340**, a target body **350**, and feedback elements **360**. The feedback loop **300** may have alternative configurations than shown in FIG. **3**, including for example different, fewer, or additional components. For example, the feedback loop **300** may include a decision point between the feedback elements **360** and the summing point **320** that determines whether to proceed to the summing point **320** or return to the preprocessing **310**. In this example, the mobility augmentation system **130** may determine, using the output of the feedback elements **360** (i.e., the feedback signal) that the user has changed his movement goal. The changed movement goal may be input into the preprocessor **310** to determine an alternative set of target movement signals.

[0090] The system **130** receives a target movement goal as an input to the feedback loop **300**. For example, the target movement goal may be received by the mobility augmentation device **120a** from the user's therapist at the remote therapy system **150**. The system **130** uses the preprocessor **310** to convert the target movement goal into a signal domain (i.e., the sensors' domain) for summation by the summing point **320**. The preprocessor **310** may have similar functionality with the preprocessing module **240**. For example, the preprocessor **310** may perform various DSP techniques such as filtering and down-sampling on the received target movement signals. The preprocessor **310** may transform a target movement goal into one or more target movement signals corresponding to the target movement goal (e.g., kinematic signals representing the movement in the movement goal).

[0091] The summing point **320** subtracts a feedback signal measured by the feedback elements **360** from the preprocessed target movement signal. The mobility augmentation system **130** then inputs

the resulting error signal into the controller **330**. The controller **330** may have similar functionality with the controller **121**. In some embodiments, the controller **330** is a proportional-integral-derivative (PID) controller that adjusts coefficients in the control function to minimize the value of the error signal it receives over time. Additionally or alternatively, the controller **330** applies any one of the responsive optimization techniques of the responsive optimization module **261**. The mobility augmentation system **130** may apply Machine Learning Control to design or tune the controller **330**. For example, the controller **330** may be a Fuzzy Logic controller that is optimized, using machine learning, based on data from multiple users or to personalize the controller for a single user.

[0092] The mobility augmentation system **130** provides the output of the controller **330** to the FES generator **340**. The system **130** uses the FES generator **340** to generate actuation signals to apply to the target body **350**. The FES generator **340** may determine, based on the output of the controller **330**, parameters for the actuation signals such as the amplitude, frequency, pulse width, or pulse shape of the actuation signal (i.e., the FES). The system **130** may apply the FES using actuator electrodes located at particular positions on the target body (e.g., at the gastrocnemius muscle of the right leg).

[0093] The mobility augmentation system **130** measures the movement resulting from or augmented by the FES using the feedback elements **360**. The feedback elements may include the sensors **123**, which may include an IMU sensor, an EMG sensor, a foot pressure bed, or a camera. The feedback signal is input to the summing point **320** to be subtracted from a subsequent value of the target movement signal to obtain an updated error signal.

Mobility Visualization and Reporting

[0094] FIGS. **4-6** depict visualizations **400**, **600**, and **700**, respectively, of data measured by sensors of a mobility augmentation device. The visualizations **400**, **600**, and **700** may be included in a report generated by the mobility augmentation system **130** and provided to the mobility management system **110** or the remote therapy system **150**. For example, a physical therapist using the remote therapy system **150** may request a report summarizing the user's gait cycle to assess the user's recovery from a leg injury. The report may include the visualization **400** indicating the electroactivity in muscles involved in a gait cycle, the visualization **600** indicating the kinematic signals in muscles involved in the gait cycle, and the visualization **700** indicating foot pressure during the gait cycle.

[0095] The visualization **400** depicts a gait cycle represented with EMG data. The EMG data represents the activation of a user's lower leg muscles. The visualization **400** includes signals **401-403** that represents the gait cycle over time and time points **404-408** that represent specific points of clinical significance during the gait cycle, including toe off, heel strike, mid stance, and mid swing. The visualization **400** shows, for each of the signals **401-403**, the EMG signal amplitude of the gastrocnemius, one of the calf muscles, as a percentage of the average EMG signal amplitude over a gait cycle. The x-axis shows percent of gait cycle where 0% is a heel strike and 100% is the moment just before the next heel strike.

[0096] The EMG signal across multiple steps may be averaged together to produce the signals **401-403** representing the mean EMG profile across those steps. For example, an RMS is applied to the EMG data of each muscle and then normalized by the mean EMG RMS value during walking. The shaded area represents one standard deviation of the EMG profile across all of the steps. The signal **401** represents an averaged EMG signal measured at the left calf's gastrocnemius muscle. The signal **402** represents an averaged EMG signal measured at a more medial location of the left calf's gastrocnemius muscle. The signal **403** represents an averaged EMG signal associated with the left tibialis anterior and measured at a muscle on the front of the lower leg.

[0097] In some embodiments, the visualization **400** may be used as a template for determining actuation based on muscle firing events identified in the visualization **400**. A machine-learned model of the activity prediction module **270** may identify the user's intended movement is a gait

based on a set of presently-measured EMG signals measured from the same positions on the user's body at which the EMG signals **401-403** were measured. The actuation determination module **260** may compare the presently-measured EMG signals to the EMG signals **401-403** of the template to identify muscle firing events that are mapped to respective actuation strategies. In response to identifying a muscle firing event, the module **260** may apply the corresponding actuation strategy. For example, upon identifying, based on the presently-measured EMG signals, a muscle firing event corresponding to the kinematic signal **404** associated with a toe off of the user's contralateral foot, the module **260** may apply an actuation strategy associated with ankle flexion of a contralateral foot.

[0098] In some embodiments, the visualization **400** is the product of measured data (e.g., motor intent data and movement data) and application of a machine-learned model of the activity prediction module **270**. For example, the general movement model **272** of the module **270** identifies that the motor intent data associated with the EMG signals **401-403** are indicative of a gait with 20% confidence while the gait is about 10% through its full cycle. As the user continues through his gait, the model **272** determines that the EMG signals **401-403** are indicative of a gait with increased confidence and identifies events within the gait cycle.

[0099] In some embodiments, as the general movement model **272** identifies events within the gait cycle, the confidence score associated with the EMG signals **401-403** indicating a gait increases. For example, the model **272** identifies the kinematic signal **404** associated with a toe off of the user's contralateral foot based on the EMG signals **401-403** through approximately 20% of the gait cycle. The model **272** may then identify the kinematic signal **406** associated with a user's leg in mid-stance based on the EMG signals **401-403** through approximately 35% of the gait cycle, the kinematic signal **405** associated with a heel strike of the contralateral foot based on the EMG signals **401-403** through approximately 50% of the gait cycle, the kinematic signal **408** associated with a toe off of the user's foot based on the EMG signals **401-403** through approximately 73% of the gait cycle, and the kinematic signal **407** associated with the user's leg in mid-swing based on the EMG signals **401-403** through approximately 89% of the gait cycle. In this way, the model **272** may determine the intended movement is a gait cycle with increasing confidence scores as the kinematic signals **404**, **406**, **405**, **408**, and **407** of the visualization **400** are identified over time.

[0100] FIG. 5 depicts a visualization **500** of kinematics of muscles involved in a user's gait. The kinematics represent the knee joint angle during a gait cycle. The kinematic signals **501-503** represent kinematics associated with flexion, varus or valgus, and rotation. The x-axis shows a percentage of gait cycle where 0% is heel strike and 100% is a moment just before the next heel strike. The kinematics (i.e., the knee joint angle) are measured across multiple steps to produce the data shown in the visualization **500** of an average gait cycle. The signal **501** represents the average knee flexion angle, the signal **502** represents the average knee varus or valgus, and the signal **503** represents the average knee rotation during the average gait cycle. The time points **504-508** are represented as vertical lines that demarcate typical points of clinical significant during the gait cycle including toe off, mid stance, and mid swing as well as toe off of the contralateral leg and heel strike of the contralateral leg.

[0101] FIG. 6 depicts the visualization **600** of foot pressure reported by the mobility augmentation system **130** during a user's gait. The sensors **123** may include a foot pressure bed. The mobility system **130** may use movement data measured at the foot pressure bed to obtain the visualization **600**. The visualization **600** shows foot pressure normalized by the maximum pressure value during the measurement. Alternatively, the mobility augmentation system **130** may normalize the foot pressure data using the mean foot pressure values during an activity, the maximum foot pressure during an activity (e.g., during quiet standing), or any suitable measurement of foot pressure. Although depicted as normalized, the visualization **600** may also be represented by un-normalized units of pressure (e.g., Pascals). At the instance of time shown in the visualization **600**, the user is striking his heel to the ground, causing a greater pressure at the heel than at the sole. Just as the

actuation determination module **260** can use the data of the visualization **400** to identify a muscle firing event within a movement and determine a corresponding actuation strategy, the module **260** may also use the data of the visualization **600** to determine an appropriate actuation strategy. For example, the maximum pressure at the heels may be indicative of a heel strike within a gait and the module **260** may apply an actuation strategy for a heel strike.

[0102] In some embodiments, the visualization **600** is the product of measured data (e.g., movement data) and application of a machine-learned model of the activity prediction module **270**. For example, the general movement model **272** of the module **270** identifies that the movement data measured using a foot pressure bed is indicative of a heel strike of a gait cycle with 70% confidence. The machine learning model training engine **271** may use visualizations such as the visualization **600** to train the model **272** to identify events at the feet such as a toe off or heel strike. [0103] In some embodiments, as the general movement model **272** identifies events within the gait cycle, the confidence score associated with the movement data increases, where the movement data indicates a gait and is measured using the foot pressure bed. Although not shown, additional visualizations from data measured using the foot pressure bed over time may be used to identify various kinematic signals associated with a gait throughout a gait cycle. Similar to the intended movement identification process as described in the description of the visualization **400**, the model **272** may determine the intended movement is a gait cycle with increasing confidence scores as the foot pressure bed data over time, including the visualization **600**, are analyzed over time.

Examples of Mobility Augmentation Devices

[0104] FIGS. 7-9 illustrate various wearable form factors of mobility augmentation devices such as the mobility augmentation device **120a** or **120b** of FIG. 1. Form factors are structured to enable personalized delivery of movement augmentation, optimize the comfort of wearing the mobility augmentation devices, and increase the efficacy of the movement augmentation.

[0105] For example, the form factors can simultaneously enable personalized delivery and increase the efficacy of movement augmentation through an array-based approach. That is, the form factor allows for multiple mobility augmentation devices to be placed around various locations of the body. After the devices are initially positioned, the mobility augmentation system may apply calibration (e.g., an actuation type of amplification) to determine the user's sensitivity to the actuation signals (e.g., FES signals) at the initial positions. The mobility augmentation system may determine that a device is positioned at a location that is not conducive to applying augmentation. For example, the user may position a device's actuating electrode closer to a bone (e.g., the ankle bone) than to a muscle (e.g., the soleus muscle of the ankle), diminishing the effects of FES actuation intended for muscle rather than bone. The mobility augmentation system may provide an indication to a user that the device placement should be adjusted.

[0106] In some embodiments, the mobility augmentation system may determine directions to provide to the user for adjusting the positioning of a mobility augmentation device. The mobility augmentation system may determine the locations of a user's mobility augmentation devices relative to one another as they are initially placed on the user's body. For example, the mobility augmentation system may use wireless communications circuitry on each device to transmit or receive radio frequency (RF) signals and determine, based on the time between transmission and receipt of an RF signal, the distance between two devices. The mobility augmentation system may determine a device is improperly positioned if, after determining a first device is properly positioned during calibration, a second device is not within a threshold range of distances from the first device, the second device's calibration results indicate poor actuation delivery, or a combination thereof. The threshold range may depend upon the target location on the body at which the device is intended to measure movement and motor intent data. For example, the user may specify, prior to positioning a mobility augmentation device, that the first device is intended to be placed at the knee and the second device is intended to be placed near the ankle.

[0107] FIGS. 7A and 7B illustrate a wearable form factor **700** of a mobility augmentation device

for a user's legs. The form factor **700** may be a pair of leggings that are outfitted with multiple sensors and actuators. For example, electrodes **741a-d** may be either sensing electrodes to measure the electroactivity at the respective locations on the user's body or actuating electrodes to apply the actuation strategy at the respective locations.

[0108] FIGS. **8A** and **8B** illustrate wearable, modular form factors **820** and **821** of respective mobility augmentation devices. The devices may have the same, initial form factor and be adjusted to be placed around different locations of the user's body. For example, the device with the form factor **820** is adjusted to be placed around a calf while the device with the form factor **821** is adjusted to be placed around a thigh. The modularity of the mobility augmentation devices allows the user to adjust the position of each mobility augmentation device to optimize the comfort of wearing the mobility augmentation devices. For example, a device may have a belt, Velcro strap, stretch material, any other suitable adjustable mechanism for increasing tightness around a limb to maintain position, or a combination thereof. In this way, the modular mobility augmentation devices are adjustable to be positioned comfortably at a location or fit of the user's choosing.

[0109] FIGS. **9A** and **9B** illustrate a wearable form factor **900** of a mobility augmentation device for a user's foot. The wearable form factor **900** is depicted as having a foot pressure bed sensor **910** that may be inserted and removed from the user's existing shoes or embedded into a pair of shoes that serve as the mobility augmentation device. The foot pressure bed sensor **910** may be coupled to one or more processors **920** (e.g., processors performing the functions of the mobility augmentation system **130**). In one example, the mobility augmentation device **120a** has the wearable form factor **700** and the mobility augmentation device **120b** has the wearable form factor **900**. In this way, the combination of the devices **120a** and **120b** can measure and apply stimulation for augmenting movement involving the legs, feet, or combination thereof.

Processes for Applying and Optimizing Mobility Augmentation Systems

[0110] FIG. **10** is a flowchart illustrating a process **1000** for applying actuation signals to an augmentation device worn by a user. In some embodiments, the mobility augmentation system **130** performs operations of the process **1000** in parallel or in different orders, or performs different steps. For example, in addition to or as an alternative to determining **1003** actuation signals and applying **1004** the signals, the process **1000** may determine, based on the identified difference, a reward or penalty for an actuation strategy that contributed to the monitored **1001** movement signals. For example, if the identified **1002** difference is small, the system **130** may reward the actuation strategy used. In this way, the process **1000** may be used to optimize how the mobility system **130** applies actuation strategies.

[0111] The mobility augmentation system **130** monitors **1001** movement signals representative of movement of a user. The system **130** is coupled to the sensors **123** that may continuously measure and provide the user's movement signals, enabling the system **130** to monitor the user's movement signals. For example, the sensors **123** include IMU sensors for measuring kinematic and kinematic data that represent the user's movement.

[0112] The mobility augmentation system **130** identifies **1002**, using a machine-learned model configured to determine an intended movement, a difference between the movement signals and target movement signals representative of the intended movement. The system **130** uses a machine-learned model of the activity prediction module **270** to determine the intended movement. For example, the general movement model **272** determines, based on EMG signals measured by one or more of the sensors **123**, that the user intends to make a gripping movement. Using the determined intended movement, the system **130**'s actuation determination module **260** determines an actuation strategy corresponding to an intention to grip an object, where the actuation strategy includes target movement signals. The target movement signals may be specified by the mobility management system **110** and be a representation of a neurotypical population performing the intended movement. The system **130** then determines a difference between the monitored **1001** movement signals and the target movement signals (e.g., the summing point **320** of the feedback loop **300**).

[0113] The mobility augmentation system **130** determines **1003** actuation signals based on the identified **1002** difference. In some embodiments, the system **130** modifies parameters of previously applied actuation signals based on the identified **1002** difference to determine **1003** updated actuation signals. For example, the identified **1002** difference may be smaller than a previously identified difference and the system **130**, in response to this decreasing difference, may lessen the intensity of the current amplitude of the FES actuation signals. Alternatively or additionally, the system **130** may access predefined actuation signals that are mapped to particular actuation strategies and identified differences.

[0114] The mobility augmentation system **130** applies **1004** the actuation signals to the one or more mobility augmentation devices worn by the user. In some embodiments, the system **130** uses the actuators **122** to apply the determined **1003** actuation signals to target locations on the user's body. For example, the system **130** may be communicatively coupled to mobility augmentation devices having modular form factors like the form factors **820** and **821**. The system **130** on one of the devices may use a FES generator (e.g., the FES generator **340**) to generate the actuation signals and apply the signals using that device's actuators. The system **130** may then transmit instructions to another device on the user's body for generating and applying the actuation signals using the other device's onboard FES generator and actuators.

[0115] FIG. **11** is flowchart illustrating a process **1100** for training a machine learning model configured to output a movement prediction based on monitored motor intent data. In some embodiments, the mobility augmentation system **130** performs operations of the process **1100** in parallel or in different orders, or may perform different steps. For example, labeling **1102** the motor intent data may be preceded by determining a label for the motor intent data if the label for the collected **1101** motor intent data is not prespecified (e.g., by the mobility management system **110**).

[0116] The mobility augmentation system **130** collects **1101** motor intent data of one or more users from a database. The motor intent data may include EMG data, IMU data, foot plantar pressure signals, kinetic signals, or a combination thereof. The one or more users may have neurotypical or neuro-atypical movement. The motor intent data collected from neurotypical users may be referred to as neurotypical motor intent data while the motor intent data collected from neuro-atypical users may be referred to as neuro-atypical motor intent data. The motor intent data may represent general movements or a specific gesture (e.g., a step, grasp, lift, or contraction). The motor intent data may represent various movements performed by a target user. For example, the motor intent data can capture how the target user's left leg moves.

[0117] The mobility augmentation system **130** labels **1102** the motor intent data with an intent label representative of intended motion characterized by the motor intent data. The intent label determination module **230** of the system **130** may use computer vision to derive labels for the motor intent data. For example, the motor intent data includes videos of the one or more users performing a gesture (e.g., a step backwards) and the intent label determination module **230** uses computer vision to determine that videos share a common pattern representative of a user taking a step backwards.

[0118] The mobility augmentation system **130** creates **1103** a first training set based on the labeled motor intent data. The machine learning model training engine **271** of the mobility augmentation system **130** may generate a set of feature vectors from the motor intent data associated with the label for taking a step backwards. The feature vectors may represent various data types such as EMG, foot pressure, and IMU signals associated with the one or more users taking a step backwards. The machine learning model training engine **271** may then use the feature vectors and the label to train a machine learning model (e.g., determining a set of weights for the machine learning model).

[0119] The mobility augmentation system **130** trains **1104** a machine learning model using the first training set, where the machine learning model is configured to output a movement prediction. The machine learning model may identify general movements when trained **1104** using the collected

1101 motor intent data associated with the users' general movements. The machine learning model may determine a movement prediction for various actuation types. For example, with the contralateral replay actuation type, the movement prediction may correspond to target movement signals that are performed by the user at another location of the user's body (e.g., a contralateral leg having neurotypical movement). That is, the system **130** may receive the target movement signals from another mobility augmentation device on the user and the activity prediction module **270** may identify the movement represented by the target movement signals.

[0120] The mobility augmentation system **130** creates **1105** a second training set based on the movement prediction and labeled motor intent data representative of movement signals of a target user. The movement signals of the target user may include one or more of kinematic signals, foot plantar pressure signals, kinetic signals, or a combination thereof. The system **130** may create **1105** the second training set similarly to how the first training set was created **1103**. For example, the system **130** generates a set of feature vectors from the motor intent data associated with the user's movement signals and the movement prediction of taking a step backwards to create the second training set.

[0121] The mobility augmentation system **130** re-trains **1106** the machine learning model using the second training set such that the machine learning model is optimized to motions of the target user. In some embodiments, re-training **1106** includes determining a similarity score between the user's movement signals and target motion and adjusting a strength of association between the monitored motor intent data and the movement prediction associated with the target motion. For example, the system **130** may strengthen an association between monitored motor intent data and the movement prediction in response to determining that a similarity score computed based on target movement signals and the user's movement signals exceeds a threshold. In another example, the system **130** may weaken the association between the monitored motor intent data and the movement prediction in response to determining that the similarity score failed to exceed the threshold. In some embodiments, the similarity score may indicate a degree of symmetry in the user's gait (i.e., the degree to which the left and right legs move similarly while walking).

[0122] Similarly, re-training **1106** may include adjusting a strength of association between the monitored motor intent data and the movement prediction associated with the target motion in response to identifying a wanted or unwanted movement feature in the user's movement signals. For example, the system **130** may strengthen an association between monitored motor intent data and the movement prediction in response to detecting that the user's toes turned upward during a gait, which is a neurotypical movement within a gait cycle. The system **130** may weaken an association between monitored motor intent data and the movement prediction in response to detecting that the user's toes failed to turn upward during the gait (i.e., at least one toe dragged as the user attempted the gait).

Additional Applications of Movement Augmentation

[0123] The mobility augmentation device and system described herein may improve disease management for patients of Parkinson's Disease (PD). Clinicians currently use standardized rating scales to assess a patient's condition, track disease progression, and evaluate responsiveness to treatment. The motor section of the Unified Parkinson's Disease Rating Scale (UPDRS), which is the gold standard for assessing a PD patients' motor symptoms, requires the patient to perform a series of motor tasks, which are visually evaluated by a trained personnel. Unfortunately, these rating scales are limited by poor temporal resolution and the subjective nature of scoring. Because symptom severity fluctuates throughout the day and can worsen with multitasking, clinical exams are also unlikely to capture the real-world severity of a patient's disease. The mobility augmentation device and system described herein can be used to overcome a patient's motor impairments caused by PD and optimize the patient's drug regimen for PD.

[0124] A mobility augmentation device can predict the onset of motor symptoms of PD before they occur. The device can generate an alert to the patient or augment the patient's movement to help

manage PD. In some embodiments, the mobility augmentation device applies FES to help the patient overcome motor impairments such as gait freezes and tremors. The mobility augmentation device may be a four-channel FES array that activates different muscles. The location and duration of the FES can be customized to the patient's needs. The mobility augmentation device can be used to supplement gastrocnemius activation using EMG-triggered FES. With the addition of FES, the patient's gait may be restored to a neurotypical gait, overcoming impairments like gait freezes. Experimental results from applying arrayed FES stimulation is further described in the description of FIG. 14.

[0125] Pharmacological agents or chemical stimuli such as levodopa (l-DOPA) are used for the treatment of PD. The delivery of l-DOPA in a patient's drug regimen may be optimized using the mobility augmentation device and system described herein. For electronically assisted l-DOPA (eDOPA), the mobility augmentation device quantifies the patient's UPDRS score and monitors minute-to-minute tracking of motor symptoms. For example, EMG and IMU data may be continuously tracked for multiple hours, identifying tremors, freezing, and bradykinesia (i.e., slowness of movement) experienced by the patient. The mobility augmentation device can then report the monitored information to the patient's physician, enabling more timely and precise adjustments of l-DOPA.

[0126] In some embodiments, the mobility augmentation system may determine the efficacy of chemical stimulus like l-DOPA. The system may identify a physical condition of a patient and the chemical stimulus administered to the user to augment the physical condition. For example, the system uses EMG and IMU sensors to identify tremors in the patient. The system can access medical records (e.g., from a remote therapy system) to identify a drug (e.g., l-DOPA) administered to the patient that treats the identified tremors. The system may monitor motor intent data representative of the patient's intended movement, where the motor intent data is indicative of an efficacy of the chemical stimulus. For example, the system monitors EMG signals associated with the patient intended to grip an object in his hand, where the person's movement to make the grip is affected by the l-DOPA he is taking.

[0127] The mobility augmentation system may also monitor movement signals that represent the patient's movement. For example, IMU sensors outfitted in a mobility augmentation device that the patient is wearing over his hand (e.g., a glove), measure the kinematic and kinetic signals of the patient's fingers and palm as he grips the object in his hand. The system may use a machine-learned model (e.g., the task-specific model 273) to identify a difference between the monitored movement signals and target movement signals associated with gripping an object. Based on this identified difference, the system may determine the efficacy of the chemical stimulus. For example, the system may determine that the chemical stimulus is not augmenting the user's movement in response to determining that the magnitude of the identified difference exceeds a threshold magnitude. For example, the patient's tremors detected from an IMU sensor may be associated with an identified difference that exceeds a threshold associated with minor tremors, indicating that the drug regimen needs evaluation or adjustment.

Experimental Findings

[0128] FIG. 12 shows an experimental finding of action prediction using inertial measurement unit data. A graph 1210 shows IMU data over time and a graph 1220 shows the prediction confidence over time of each of multiple potential movement predictions corresponding to the IMU data. The movement predictions in the graph 1220 include a side step prediction 1221, a forward step prediction 1222, a backward step prediction 1223, and a center step prediction 1224. The IMU data of the graph 1210 was measured by IMU sensors of the mobility augmentation device described herein. The probabilities of the graph 1220 were determined by the mobility augmentation system described herein. The system detects a user's foot lifting at a time 1211a and determines the side step prediction 1221 at a time 1212b with approximately 100% confidence based on the measured IMU data. The system detects a user's foot lifting at a time 1211b and determines the forward step

prediction **1222** at a time **1212b** with approximately 100% confidence based on the measured IMU data. The system detects a user's foot lifting at a time **1211c** and determines the backward step prediction **1223** at a time **1212c** with approximately 100% confidence based on the measured IMU data.

[0129] FIG. **13** shows an experimental finding of action prediction using electromyography data. A graph **1310** shows EMG data over time and a graph **1320** shows the prediction confidence over time of each of multiple potential movement predictions corresponding to the EMG data. The movement predictions in the graph **1320** include a side step prediction **1321**, a forward step prediction **1322**, a backward step prediction **1323**, and a center step prediction **1324**. The EMG data of the graph **1310** was measured by EMG sensors of the mobility augmentation device described herein. The probabilities of the graph **1320** were determined by the mobility augmentation system described herein. The system determines the side step prediction **1321** at a time **1312a** with approximately 100% confidence based on the measured EMG data and detects a user's foot lifting at a time **1311a**. The system determines the forward step prediction **1322** at a time **1312b** with approximately 100% confidence based on the measured EMG data and detects a user's foot lifting at a time **1311b**. The system determines the backward step prediction **1323** at a time **1312c** with approximately 100% confidence based on the measured EMG data and detects a user's foot lifting at a time **1311c**.

[0130] While the times **1212b-c** that the IMU predictions shown in FIG. **12** occur after the user's actual movement, the times **1312a-c** that the EMG predictions of FIG. **13** occur are earlier than the user's actual movement. Specifically, the IMU predictions occurred approximately 103 milliseconds after the user's foot lifted off of the ground and the EMG predictions occurred approximately 269 milliseconds before the user's foot lifted off of the ground. The earlier predictions allow the actuation strategies to be applied earlier, as needed, reducing the likelihood that the actuation strategy will be ineffective because it is not timely applied according to the user's current movements being augmented. Hence, the mobility augmentation devices using EMG sensors may improve movement augmentation over solely using IMU sensors.

[0131] FIG. **14** shows an experimental finding of knee and hip kinematics augmented with functional electrical stimulation. A 10 year-old child with primarily unilateral spastic cerebral palsy (CP) wore the mobility augmentation devices described herein and her movements were monitored with and without FES. Her right leg showed neurotypical movement while her left leg's movement was impacted by CP. The mobility augmentation devices were positioned at each of her legs and the sensors of the devices measured kinematic signals from her gait cycle.

[0132] Graph **1400a** shows her knee kinematics without FES applied. Kinematic signal **1401a** of the graph **1400a** shows her right knee's flexion angle exhibiting neurotypical movement. Kinematic signal **1402a** of the graph **1400a** shows her left knee's flexion angle exhibiting neuro-atypical movement. Graph **1400b** shows her knee kinematics with FES applied. Kinematic signal **1401b** of the graph **1400b** shows her right knee's flexion angle exhibiting neurotypical movement. Kinematic signal **1402b** of the graph **1400b** shows her left knee's flexion angle augmented by FES and exhibiting movement that more closely aligns with the neurotypical kinematic signal **1401b** than with the neuro-atypical kinematic signal **1402a**.

[0133] Graph **1410a** shows her hip kinematics without FES applied. Kinematic signal **1411a** of the graph **1410a** shows her right hip's flexion angle exhibiting neurotypical movement. Kinematic signal **1412a** of the graph **1410a** shows her left hip's flexion angle exhibiting neuro-atypical movement. Graph **1410b** shows her hip kinematics with FES applied. Kinematic signal **1411b** of the graph **1410b** shows her right hip's flexion angle exhibiting neurotypical movement. Kinematic signal **1412b** of the graph **1410b** shows her left hip's flexion angle augmented by FES and exhibiting movement that more closely aligns with the neurotypical kinematic signal **1411b** than with the neuro-atypical kinematic signal **1412a**.

Additional Considerations

[0134] The foregoing description of the embodiments has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the patent rights to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.

[0135] Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm may be a sequence of operations leading to a desired result. These operations, while described functionally, computationally, or logically, are understood to be implemented by computer programs or equivalent electrical circuits, microcode, or the like. Furthermore, it has also proven convenient at times, to refer to these arrangements of operations as modules, without loss of generality. The described operations and their associated modules may be embodied in software, firmware, hardware, or any combinations thereof. The operations are those requiring physical manipulations of physical quantities. Such quantities may take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. Such signals may be referred to as bits, values, elements, symbols, characters, terms, numbers, or the like.

[0136] It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the present disclosure, it is appreciated that throughout the description, certain terms refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage devices.

[0137] The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the intended purposes, or it may include a computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, such as, but not limited to, any type of disk including floppy disks, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

[0138] The algorithms and displays presented herein are not inherently related to any particular computer or other apparatus. Various other systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the method. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

[0139] The present disclosure may be provided as a computer program product, or software, that may include a machine-readable medium having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory devices, etc.

[0140] Any of the steps, operations, or processes described herein may be performed or implemented with one or more hardware or software modules, alone or in combination with other devices. In one embodiment, a software module is implemented with a computer program product

comprising a computer-readable medium containing computer program code, which can be executed by a computer processor for performing any or all of the steps, operations, or processes described.

[0141] Embodiments may also relate to an apparatus for performing the operations herein. This apparatus may be specially constructed for the required purposes, and/or it may comprise a general-purpose computing device selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, tangible computer readable storage medium, or any type of media suitable for storing electronic instructions, which may be coupled to a computer system bus. Furthermore, any computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs for increased computing capability.

[0142] Embodiments may also relate to a product that is produced by a computing process described herein. Such a product may comprise information resulting from a computing process, where the information is stored on a non-transitory, tangible computer readable storage medium and may include any embodiment of a computer program product or other data combination described herein.

[0143] As used herein any reference to “one embodiment” or “an embodiment” means that a particular element, feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. The appearances of the phrase “in one embodiment” in various places in the specification are not necessarily all referring to the same embodiment. Where values are described as “approximate” or “substantially” (or their derivatives), such values should be construed as accurate $\pm 10\%$ unless another meaning is apparent from the context. From example, “approximately ten” should be understood to mean “in a range from nine to eleven.”

[0144] As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive or and not to an exclusive or. For example, a condition A or B is satisfied by any one of the following: A is true (or present) and B is false (or not present), A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0145] In addition, use of the “a” or “an” are employed to describe elements and components of the embodiments herein. This is done merely for convenience and to give a general sense of the invention. This description should be read to include one or at least one and the singular also includes the plural unless it is obvious that it is meant otherwise.

[0146] Finally, the language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the patent rights. It is therefore intended that the scope of the patent rights be limited not by this detailed description, but rather by any claims that issue on an application based hereon.

Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the patent rights, which is set forth in the following claims.

Claims

1. A method comprising: retrieving a set of previously determined movement predictions associated with a target user and corresponding previous actuation instructions applied based on each movement prediction; labeling the set of previously determined movement predictions based on measurements taken during application of respective previous actuation instructions to generate a training set; and training a machine learning model using the training set, the machine learning model configured to output, based on monitored motor intent data of the target user, a movement

- prediction corresponding to likely motion characterized by the monitored motor intent data.
2. The method of claim 1, wherein the monitored motor intent data comprises one or more of electromyography (EMG) data, inertial measurement unit (IMU) data, foot plantar pressure signals, or kinetic signals.
 3. The method of claim 1, wherein the measurements taken during application of respective previous actuation instructions include one or more of foot plantar pressure signals, kinematic signals, and kinetic signals.
 4. The method of claim 3, further comprising: tracking an amplitude of a signal of the measured movement; and comparing the tracked amplitude to a baseline signal profile of the movement.
 5. The method of claim 1, wherein the machine learning model is further configured to output, based on the monitored motor intent data, an actuation instruction corresponding to the likely motion characterized by the monitored motor intent data.
 6. The method of claim 5, wherein the actuation instruction includes a proportional steering ratio between electrical signals transmitted at electrodes of a wearable stimulation array.
 7. The method of claim 5, further comprising: stimulating movement of the target user using the actuation instruction; determining a level of fatigue experienced by the target user during the stimulation; and adjusting the actuation instruction based on the level of fatigue.
 8. The method of claim 7, wherein determining the level of fatigue comprises: measuring a first set of EMG signals of the target user during the stimulation; determining a frequency response of electroactivity in the first set of EMG signals; and comparing the frequency response to a baseline frequency response determined using a second set of EMG signals measured when the user was rested.
 9. The method of claim 1, wherein the movement prediction corresponds to a movement template representative of a sequence of muscle firing events, the sequence of muscle firing events associated with a plurality of target movement signals.
 10. The method of claim 1, wherein the movement prediction corresponding to the likely motion comprises a likely IMU data value at a subsequent time occurring after a time at which the monitored motor intent data is monitored.
 11. The method of claim 1, wherein generating the training set further comprises determining a feature vector representative of muscle firing events located at different areas of the target user's body.
 12. The method of claim 1, wherein the machine learning model is trained using one or more processors of a remote device.
 13. The method of claim 1, wherein training the machine learning model using the training set occurs during a second stage of training and wherein the machine learning model generates the previously determined movement predictions after a first stage of training.
 14. The method of claim 13, wherein the first stage of training comprises training the machine learning model using a set of labeled motor intent data collected across one or more users.
 15. The method of claim 14, wherein the set of motor intent data is labeled with an intent label representative of intended motion characterized by the set of motor intent data.
 16. The method of claim 15, wherein the set of motor intent data is labeled by one or more of: the one or more users, one or more observers watching the one or more users, a computer vision algorithm configured to recognize motions within video, or a system configured to prompt the one or more users to perform particular actions.
 17. The method of claim 15, further comprising determining the intent label based on one or more of foot plantar pressure signals, kinematic signals, and kinetic signals of one or more users from a database.
 18. The method of claim 15, wherein the motor intent data comprises one or more of neurotypical motor intent data or neuro-atypical motor intent data.
 19. A mobility improvement system comprising a non-transitory computer-readable storage

medium storing instructions for execution and a hardware processor configured to execute the instructions, the instructions, when executed, cause the hardware processor to perform steps comprising: retrieving a set of previously determined movement predictions associated with a target user and corresponding previous actuation instructions applied based on each movement prediction; labeling the set of previously determined movement predictions based on measurements taken during application of respective previous actuation instructions to generate a training set; and training a machine learning model using the training set, the machine learning model configured to output, based on monitored motor intent data of the target user, a movement prediction corresponding to likely motion characterized by the monitored motor intent data.

20. A non-transitory computer readable storage medium storing executable instructions that, when executed by one or more processors, cause the one or more processors to perform steps comprising: retrieving a set of previously determined movement predictions associated with a target user and corresponding previous actuation instructions applied based on each movement prediction; labeling the set of previously determined movement predictions based on measurements taken during application of respective previous actuation instructions to generate a training set; and training a machine learning model using the training set, the machine learning model configured to output, based on monitored motor intent data of the target user, a movement prediction corresponding to likely motion characterized by the monitored motor intent data.
